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UNIVERSITY OF PORT HARCOURT

CHM 121 — Past Questions & Materials

by Renance

Course: CHM 121

Institution: University of Port Harcourt

Papers in this pack: 1

Levels: 100

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Paper 1 of 1 — CHM 121 foundation chemistry 1

Level 100 • First Semester • PastQuestion

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____—____ FACULTY OF SCIENCE

a DEPARTMENT OF CHEMISTRY

2016/2017 FIRST SEMESTER EXAMINATION APRIL 2017

CHM 121; FOUNDATION CHEMISTRY I UNIT: 3 TIME ALLOWED:.2 HRS

INSTRUCTIONS:

(a) Attempt all questions.

(b) Write the correct answers ONLY

(c) Constants: $1 \text{ nm} = 10^9 \text{ m}$, $h = 6.625 \times 10^{-34} \text{ Js}$ and $c = 3 \times 10^8 \text{ m/s}$

(d) $H=1$, $C=12$, $O=16$, $Na = 23$, $N=14$, $S=32$

The evidence that electrons were discrete particles was obtained by

In cathode tube experiment, when a high voltage is applied to any gas at low pressure, streams of particles

moved from__to

Write a balanced equation for the reaction of oxalic acid and sodium hydroxide.

20 cm³ of hydrochloric acid was titrated against 25 cm³ of sodium hydroxide solution that contained 1 dm³, Calculate: (i) concentration of A in mol/dm³ (ii) concentration of A in g/dm³.

is the apparatus for transferring solid substances.

The apparatus used for drying are__—_ and __

Rutherford in his atomic model experiment discovered

What is the mass of an electron

theory postulated that elements are made up of indivisible particles called atoms.

. The quanta of energy emitted by hot bodies are given by__-

. Degenerate orbitals are orbitals of __ energies.

_ Covalent bonds are formed as a result of

_ An element with atomic number 14 will have __

_ A sublevel with azimuthal quantum number, $l = 3$ will have _ orbitals.

. Elements in periods 6 and 7 are called __ and are called __.

. The scattering of alpha particles is mainly caused —

- The law of definite proportion states that: _____

_ Write the electron configuration of S²⁻.

. The theory that atom consists of a central nucleus and orbiting electrons was based on

_ Mention one characteristic common to elements in the same period.

_ In the periodic table, elements are arranged according to their_____

. The tendency of an atom to lose electron when energized is_

- . The energy emitted or absorbed by an element is said to be in specific amounts called ____.
- . How many electrons are there in the Lewis structure of NO_3^- ?
- _ A spectrum in which different colours merge into each other is called ____.
- _ What is the wavelength, in nanometers, of a photon of light whose frequency is $5.53 \times 10^{10} \text{ Hz}$?
- . What is a standard solution?
- _ Covalent compounds have predominantly lower melting and boiling points because ____.
- _ Give the number of lattice points in a body centered cubic unit cell.
- . Electrical and thermal conductivity ____ across the period and ____ down the group.
- . Bohr's theory of atom was inadequate in explaining ____.
- . What phenomenon does an NMR spectrum depict?
- _____ has 3.60g in _____.
- _____ principal quantum number.
- _____ Wavelength 800 nm.
- . Which groups of elements are called the p block elements. List the elements.
- _____ is Co.
- . How many unpaired electrons are there in Fe^{2+} ($Z = 26$)?
- . Give the sublevel name of quantum number, $n = 5, l = 2$.
- . Arrange the following in order of increasing ionic radius; K^+ , Cl^- and P^{3-} .
- . Mention the bond types in NH_3 and C_2H_2 .
- . In what group and period does element with atomic number 19 belong.
- . Write the product of a compound formed by the combination of an element A having atomic number 12 with element B of atomic number 9.
- . Calculate the molar mass of sodium sulphate.
- . What is the charge on the atom with 12 protons, 11 neutrons and 10 electrons?
- . The regular arrangement of ions in solid crystals is called ____.
- . Which is the highest occupied energy orbital in a silicon atom?
- . Mention the number of electrons that can be accommodated by the quantum number, $n = 5$?
- . Calculate the wavelength of radiation that emitted $3.31 \times 10^{-19} \text{ J}$.
- . Which principal energy level will contain electrons with the lowest energy?
- . How many orbitals are contained in the second principal level ($n=2$) of a given atom?
- . The s block elements of the periodic table are made up of ____ and ____.
- . Which of the rules does each of the following electronic energy diagram violate?
- _____ is the correct answer.
- . How many sublevels are there in the third principal quantum number?
- . What type of bond exists in diatomic phosphorus?
- . Arrange the following in order of decreasing atomic size; F, Ne and Mg.

- . Mention any factor that affects ionization energy.
- limits the maximum number of electrons in an orbital to two.
- . The scientific proof that cathode rays are negatively charged particles was made by
- . The periodic table is divided into horizontal rows known as__and vertical columns known as __
- . Arrange the following in order of decreasing electronegativity; N, Si and Al.
- ___arises because of small instantaneous dipoles in atoms.
- . Octet rule refers tothe ___ during chemical combination.
- . The chemical bond that holds together the atoms of solid metal or alloy is
- . The magnetic quantum number explains __of orbitals.
- . Atomic numbers were discovered by ___ while the term orbital was first coined by
- . All___ compounds are insoluble in organic solvents but soluble in polar solvents.
- . The orbital names: s,p,d and f were derived from = : and

Answer true or false for questions (65 - 70)

- . X- rays can be deflected by both magnetic and electric fields.
- . Atoms of the same elements having different atomic number but same mass number are isotopes.
- "\
- . In ionic bonding electrical forces between different charged ions hold the atoms together.
- . Beryllium has higher ionization energy than oxygen.¢
- . Line spectra cannot be used in the identification of elements. ¢
- . Atoms with larger size are more electronegative than atoms of smaller size. ¢

GOOD LUCK